

GSM where the frequencies are used with a certain buffer space. This is essentially one of the reasons that CDMA networks are easier to set up—and this also has the additional payoff of better spectral efficiency.

The BTS consists of a several transceivers that can simultaneously handle full duplex communications, and generally have towers that support several transmitting and receiving antennas. The BTS adjusts the transmitting power of the mobile phones that it services and takes care of several CDMA call processing functions. It takes the digital signal from the MTSO and encodes and interleaves it. This ensures redundancy and takes care of error correction coding. At this point, the digitised voice signal is then encoded to separate it from all the other voice signals. After this, the encoded signals are channelised or spread over the entire bandwidth of the CDMA channel. Finally, the channelised data is combined into one analogue RF (Radio Frequency) signal, which is transmitted over the air. The BTS thus serves as a bridge between all the mobile phone users in the area and connects the simultaneous calls via high-speed cable links or microwave radio links to the MTSO.

The MTSO is the nerve centre of the operations—similar to central offices in the telephonic system, the MTSO han-

dles all the control processing, and along with the switching network, maintains the updated database of subscribers and billing records. The MTSO also takes care of compressing the digitised speech signals for more efficient transmission over the scarce radio spectrum. This is known as vocoding. To communicate with other MTSOs, BTSs as well as the PSTN, the MTSOs use a switching network that makes the actual physical connection between a consumer on the radio side of the system and the consumer on the PSTN. The switching network in turn routes calls, implements all call features, as well as stores the billing records and interpreting dialled phone numbers. The MTSO or Switch also stores the Home Location Register (HLR). The HLR is the official database of all the customers on the wireless system. The HLR may be located at another more centralised location, where several switches can query it. It generally holds the information about the phone, its technical parameters, and the subscriber's account status. The HLR also keeps track of whether the phone is turned on, and if so, which switch is currently serving the phone.

When a mobile phone is switched on, it first scans through the group of preset control channels, to select the BTS that beams out the strongest signal. The mobile phone then registers with

the system to inform that it is up and ready to send and receive phone calls. The phone then monitors that control channel till the signal drops below usable strength. At this point, the mobile phone scans all the control channels in search of the strongest signal. Since the control channels are standardised throughout the entire region or country, all phones scan the same channels, whilst on standby.

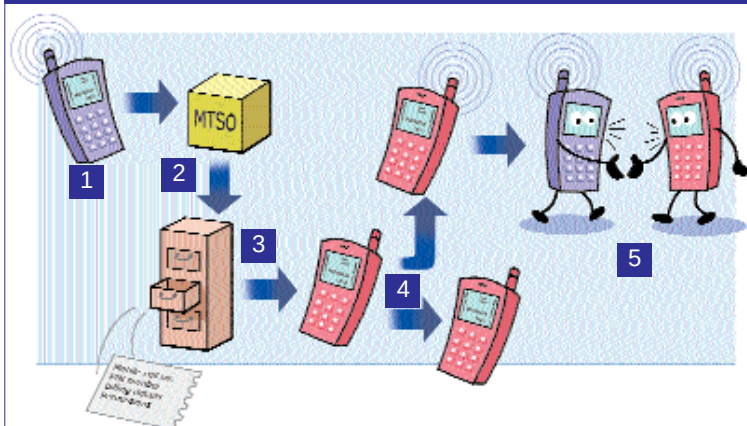
Hello, operator?

When a telephone call is placed to a mobile phone, the system checks with the HLR for the current location of the phone and pages the phone's identification number—the subscriber's telephone number—over the control channels. The phone then recognises the paged message and sends back an acknowledgement to the strongest BTS. The MTSO then instructs the BTS to move the call over to an unused voice channel, and the BTS then responds by instructing the mobile phone to move over to the correct frequency. The phone acknowledges the transmission and jumps over to the assigned voice channel. At this point of time the mobile phone alerts the subscriber of an incoming call with a ring.

When the user of a mobile phone initiates a call, the initiation request is sent to the BTS that contains the mobile

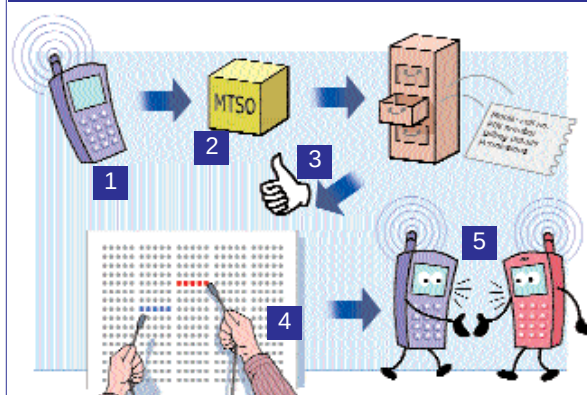
INFOGRAPHICS: Mahesh Benkar

Anatomy of an Incoming Call



- 1 The incoming signal reaches the MTSO...
- 2 The MTSO checks with the HLR for the current location of the phone...
- 3 It then sends off messages all over the control channels in the area...
- 4 The phone receives them, and if it is ready to take the call, rings and alerts the user; Else it doesn't accept the call
- 5 If the phone can take the call, the connection is set and the conversation begins

Anatomy of an Outgoing Call



- 1 The phone messages the MTSO with its own number and the target phone number
- 2 The MTSO does a validation check with the HLR
- 3 As soon as it gets the OK signal
- 4 The Switch routes the call and sets the connection
- 5 And the conversation begins